100 DAYS QUALITY MANAGEMENT CHALLENGE • DAY 91

Airport Equipment Reliability & Quality Management

Preventive Maintenance, Failure Investigation and Corrective Action

Organisation: **Airports Authority of India (AAI)** | Author: **Jagriti Singh**

1. INTRODUCTION

In modern airport operations, equipment reliability and preventive maintenance are critical to ensuring operational safety, passenger safety, and punctuality. Airport Ground Support Equipment (GSE)—such as baggage tractors, pushback tugs, passenger boarding stairs, and aircraft fueling units—operates in demanding, time-critical environments. A mechanical breakdown can disrupt flight schedules, block taxiways, or compromise safety around active aircraft.

Implementing robust quality engineering practices—such as scheduled preventive maintenance, structured failure investigations, and verified corrective actions—minimizes unexpected downtime, extends equipment operational life, and maintains strict adherence to aviation safety protocols.

2. PRACTICAL CASE STUDY (EDUCATIONAL SCENARIO)

SCENARIO: RECURRENT MECHANICAL MALFUNCTION IN GROUND SUPPORT EQUIPMENT

1. Operational Problem Identification

A fleet of airport baggage towing tractors experiences repeated hydraulic system overheating and pressure loss during peak turnaround operations. Remarkably, these failures persist even shortly after routine scheduled maintenance, causing unexpected operational delays.

2. Operational & Safety Importance

Baggage tractors transport critical luggage cargo across busy ramp zones. Hydraulic loss compromises steering and braking responsiveness, posing severe safety risks to ground crew, aircraft, and airfield infrastructure.

3. Initial Engineering Verification & Inspection

Engineers must first inspect basic mechanical parameters: fluid levels, filter cleanliness, seal integrity, and signs of external leakage. Immediate checks should confirm whether hydraulic fluid specifications match manufacturer guidelines and whether fluid contamination is present.

4. Operating Conditions & Maintenance Records Analysis

Reviewing historical logbooks helps correlate failures with specific operating conditions (e.g., extreme ambient summer temperatures, continuous engine idling during flight delays, or specific operator shifts). Maintenance records reveal whether past servicing used non-genuine seals or incorrect hydraulic oil viscosity.

5. Root Cause Investigation

A structured investigation reveals that routine maintenance involved changing oil but failed to flush thermal bypass valves. Accumulation of micro-particles caused bypass valves to stick, restricting oil flow to the heat exchanger during heavy duty cycles. Hence, a single surface symptom (fluid overheating) resulted from a secondary failure mechanism (valves stuck due to fluid contamination).

6. Corrective & Preventive Action Implementation

Corrective actions include flushing the hydraulic cooling circuit, replacing contaminated bypass valves, upgrading to high-efficiency filtration, and updating maintenance procedures to mandate valve function checks during every service interval.

7. Verification of Action Effectiveness

Effectiveness is monitored by measuring hydraulic operating temperatures across 100 operating hours under full load conditions and verifying zero reoccurrence of fluid degradation or pressure drops.

Note: Equipment affecting operational safety must strictly follow authorized technical manuals and approved standard operating procedures (SOPs). Engineers must never assume a single cause explains every recurring malfunction.

3. KEY QUALITY TOOLS IN MAINTENANCE ENGINEERING

1. Check Sheet
Purpose: A structured, standardized form designed for real-time data collection to record event frequencies, inspection results, or failure types systematic during maintenance rounds.

Case Study Example: *Mechanics use a daily pre-shift hydraulic check sheet to log oil temperature, fluid level, and visual leakage before dispatching baggage tractors to the ramp.*

2. Root Cause Analysis (RCA)
Purpose: A problem-solving methodology aimed at identifying the fundamental underlying reason for a equipment failure rather than merely treating surface symptoms.

Case Study Example: *Investigating why hydraulic oil overheated led engineers past the pump to discover clogged cooling valves caused by un-flushed debris.*

3. Five Whys
Purpose: An iterative interrogative technique used to explore the cause-and-effect relationships underlying a specific problem by repeatedly asking "Why?".

Case Study Example: *(1) Why did fluid overheat? Cooler flow was blocked. (2) Why was flow blocked? Valve stuck. (3) Why did valve stick? Debris build-up. (4) Why was debris present? Maintenance didn't flush system. (5) Why no flushing? SOP lacked valve flushing steps.*

4. Failure Mode and Effects Analysis (FMEA)
Purpose: A systematic, proactive technique for evaluating potential failure modes, assessing their risk priority (Severity, Occurrence, Detection), and identifying preventive measures.

Case Study Example: *Rating valve clogging high on Severity (loss of steering/ramp stall) prompts setting strict preventive replacement and flushing schedules for all fleet units.*

4. KEY ENGINEERING TAKEAWAYS

- **Proactive Preventive Maintenance:** Routine servicing must go beyond simple component replacement; it requires total system function validation (e.g., valves, heat exchangers, and filters).
- **Rigorous Equipment Inspection:** Pre-operational visual and functional checks catch early indicators of wear before catastrophic failure occurs on active taxiways.
- **Data-Driven Failure Investigation:** Symptom-based repairs often fail; engineers must utilize structured tools like RCA and 5 Whys to locate true systemic causes.
- **Meticulous Maintenance Records:** Complete maintenance logs provide vital historical trends for operating conditions, part replacement intervals, and repeat failure patterns.
- **Verification of Corrective Action:** A repair is not complete until post-maintenance testing under load validates that the failure mode is fully resolved without collateral operational impact.

5. TECHNICAL INTERVIEW QUESTIONS & ANSWERS

Q1: What is the main objective of Preventive Maintenance (PM) in airport Ground Support Equipment?
Ans: To maximize equipment reliability, extend service life, prevent costly operational breakdowns on active ramps, and ensure crew and flight safety by addressing wear before failure occurs.

Q2: How does Mean Time Between Failures (MTBF) help measure equipment reliability?

Ans: MTBF measures the average operational time between inherent system failures. A higher MTBF indicates superior equipment reliability and effective preventive maintenance routines.

Q3: Why should an engineer inspect maintenance logs before troubleshooting a recurrent failure?

Ans: Logs reveal historic failure trends, recently replaced components, service intervals, and operating environments, helping narrow down systemic root causes quickly.

Q4: What immediate step should a mechanical engineer take when equipment fails right after maintenance?

Ans: Verify recent service procedures, inspect newly installed parts for correct specification/fitment, check fluid levels/viscosity, and confirm whether assembly errors occurred.

Q5: How do harsh airport environmental conditions impact ground equipment reliability?

Ans: Extreme temperatures, asphalt heat radiation, dust, and heavy load cycles accelerate oil thermal degradation, rubber seal hardening, and electrical component fatigue.

Q6: What is the primary difference between corrective maintenance and breakdown repair?

Ans: Corrective maintenance addresses identified defects or partial degradation during planned checks, whereas breakdown repair occurs after unplanned functional failure halts operations.

Q7: How does a Check Sheet contribute to quality control in maintenance operations?

Ans: It provides a standardized format to capture consistent, error-free operational data, ensuring inspection steps are not missed during routine pre-flight or daily servicing.

Q8: In Root Cause Analysis, why is it dangerous to rely on a single assumed cause?

Ans: Multiple underlying factors (e.g., poor SOP, wrong lubricant, operating error) often interact. Addressing only one symptom leads to recurring equipment failure.

Q9: Explain the role of the 5 Whys technique in failure investigation.

Ans: It systematically drills down through successive layers of symptoms by repeatedly asking "Why?", exposing procedural or systemic causes behind physical component breakdowns.

Q10: What is Risk Priority Number (RPN) in FMEA, and how is it calculated?

Ans: RPN ranks risk priorities to guide preventive action. It is calculated as: *RPN = Severity (S) × Occurrence (O) × Detection (D)*.

Q11: How do you verify whether a corrective action for a mechanical failure was successful?

Ans: By testing the system under simulated/actual full-load operating conditions over a specified monitoring duration and verifying zero recurrence of the failure indicator.

Q12: Why is fluid condition monitoring essential in hydraulic equipment maintenance?

Ans: Hydraulic fluid acts as heat transfer medium and power transmitter. Contaminated or degraded fluid causes valve sticking, seal erosion, pump cavitation, and system overheating.

Q13: How does standardizing maintenance SOPs improve equipment quality management?

Ans: Standardized SOPs eliminate human variability, ensure uniform service execution across shifts, and incorporate proven troubleshooting best practices into routine tasks.

Q14: What safety protocol must precede any physical failure investigation on heavy machinery?

Ans: Strict adherence to Lockout/Tagout (LOTO) protocols, pressure relief of pneumatic/hydraulic lines, and securing moving mechanical elements to ensure technician safety.

Q15: What is the role of continuous monitoring in Quality Assurance for airport ground equipment?

Ans: Continuous monitoring tracks real-time operating metrics (vibration, temp, pressure) to detect micro-trends, allowing predictive interventions before catastrophic field failure occurs.

6. References

- Airports Authority of India (AAI) Official Portal: www.aai.aero
- ISO 9001:2015 Quality Management Systems — Requirements.
- Mobley, R. Keith. *An Introduction to Predictive Maintenance*. Elsevier / Butterworth-Heinemann.

Disclaimer: *This is an independent educational case study based on a hypothetical scenario. It does not represent an actual AAI incident, internal procedure, or official AAI publication. This work is not affiliated with or endorsed by AAI.*